

System Working Principle

Automatic Touchless Water Dispenser Using Arduino, HC-SR04, LCD, Servo Motors, and Valve/Tap Mechanism

1. Introduction

The automatic touchless water dispenser is designed to dispense water when a user places their hand near the sensor. An Arduino acts as the main controller. The HC-SR04 ultrasonic sensor detects the hand position, the LCD displays the system status, and two servo motors operate the valve or tap mechanism. This contactless operation improves convenience, hygiene, and water-use control.

2. Main Components

Component	Function
Arduino	Initializes the system, processes distance data, controls the LCD and servo motors, and manages the dispensing sequence.
HC-SR04 ultrasonic sensor	Measures the distance between the sensor and the user's hand.
16×2 LCD	Displays messages such as "Dispensing...", "Water Stopped", and standby status.
Servo Motor 1 and Servo Motor 2	Rotate the valve/tap mechanism between closed and open positions.
Valve/Tap mechanism	Controls the physical flow of water.
Power supply	Provides regulated power to the Arduino, sensor, LCD, and servos.

3. System Working Principle

- 1. Power ON:** When power is switched ON, the Arduino starts executing the program stored in its memory.
- 2. Arduino initialization:** The Arduino configures the HC-SR04 pins, LCD, and servo motors. The servos are placed at 0°, which represents the closed-valve position. The LCD shows a standby message.
- 3. Distance measurement:** The Arduino sends a trigger pulse to the HC-SR04. The sensor emits ultrasonic waves and receives the echo reflected from the user's hand. The Arduino calculates the distance from the echo time.
- 4. Hand detection:** The measured distance is continuously compared with the programmed threshold. If the hand is at a distance of 15 cm or less, the Arduino treats it as a valid dispensing request.
- 5. Dispensing indication:** After detecting the hand, the Arduino updates the LCD with "Dispensing...". This informs the user that the system has accepted the request.
- 6. Valve opening:** The Arduino commands Servo 1 and Servo 2 to rotate to 90°. Their combined movement opens the connected valve or tap mechanism.
- 7. Water flow:** Once the valve is open, water flows through the outlet. The Arduino maintains the open position for approximately 3 seconds.
- 8. Valve closing:** After the 3-second dispensing period, the Arduino commands both servos to return to 0°. This closes the valve and stops the water flow.

9. Status update and standby: The LCD displays “Water Stopped”. The system then returns to standby and resumes distance measurement for the next user request.

4. Control Flow

Power ON → Arduino initializes → HC-SR04 measures distance → Check whether hand distance ≤ 15 cm → If NO, continue monitoring → If YES, display “Dispensing...” → Move Servo 1 and Servo 2 to 90° → Open valve/tap → Allow water flow for 3 seconds → Move servos to 0° → Close valve → Display “Water Stopped” → Return to standby.

5. Operating Logic

Condition	Arduino action
Hand distance > 15 cm	Keep the valve closed, keep servos at 0° , and continue monitoring.
Hand distance ≤ 15 cm	Display “Dispensing...” and rotate both servos to 90° .
Three seconds elapsed	Return both servos to 0° and close the valve.
Dispensing complete	Display “Water Stopped” and return to standby.

6. Important Implementation Notes

- The servos should preferably use a separate adequate 5 V supply, while sharing a common ground with the Arduino.
- The HC-SR04 must be positioned so that the hand can be detected reliably without interference from the dispenser body.
- A small delay or confirmation reading can be used to prevent accidental triggering caused by noisy distance measurements.
- The 3-second interval may be changed in the Arduino program according to the required quantity of water.
- Mechanical alignment of both servos is important so that the valve opens and closes smoothly without excessive load.

Conclusion

The system provides automatic, touch-free water dispensing by combining ultrasonic hand detection with Arduino-based control. The valve opens only when a hand is detected within 15 cm and closes automatically after 3 seconds, reducing unnecessary water flow and improving hygienic operation.